

WASP-10b

R-1823

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_160929 · created 2026-08-03T04:59:04+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457660.8868 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.21 +/- 0.061
Transit depth (Rp/Rs)^2	4.4 +/- 2.55 [%]
Orbital Inclination (inc)	89.55 +/- 1.87
Airmass coefficient 1 (a1)	0.72 +/- 0.27
Airmass coefficient 2 (a2)	0.178 +/- 0.092
Scatter in the residuals of the lightcurve fit is	38.84 %
Best Comparison Star	#3 - [105, 456]
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.094 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.21 ± 0.061 vs 0.1592 (deviation 0.78σ)
Duration fit vs published	0.094 d vs 0.0946 d (deviation 0.05σ)
Mid-time O–C	-6.9 min
Depth SNR	3.2
Residual scatter	38.84 %

Light curve

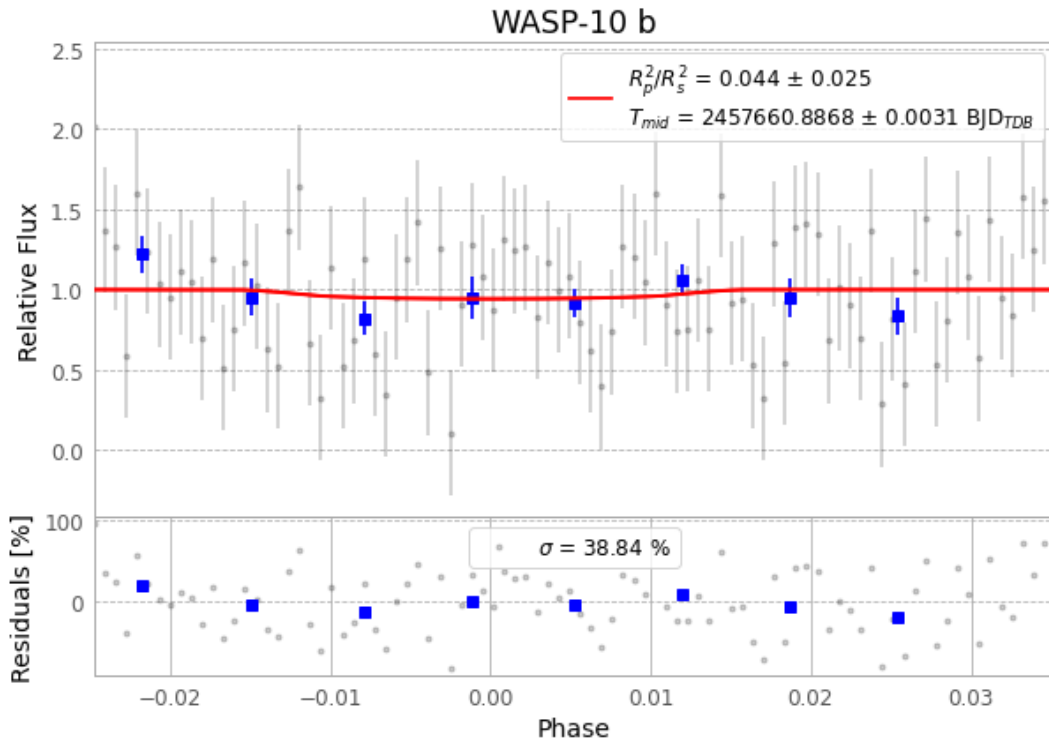


Figure 1 — FinalLightCurve_WASP-10 b_2016-09-29.png (SHA-256 440d6bf23263e677...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [306, 434], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 e646c21a1a829d7e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

90 FITS frames (59 MB), WASP-10160929071812.FITS → WASP-10160929114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-160929.log (1e8ca760b0aa...), FinalLightCurve_WASP-10 b_2016-09-29.png (440d6bf23263...), FinalLightCurve_WASP-10 b_2016-09-29.csv (f19bc37dd249...)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:59Z.